

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing

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ClothMate - A Home-autonomous system.

Proposal Document

Group 1.2/18

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Background

With modern advancements, people are usually occupied in their careers which leaves no room for domestic chores. The task of drying clothes is largely manual and weather dependent. Due to unpredictable weather conditions, people struggle with inefficient drying and increased manual effort. Hence, we propose an Arduino-based cloth line automation system that involves monitoring the rainfall and sunlight with the implementation of sensors and indicators within the system to capture data in real time. The primary goal of this system is to control the cloth line automatically when a motion of rainfall is detected by the sensors. With the reliability of this home automation system, it would be convenient for people to attend to their other duties and increase efficiency in household chores.

Problem and Motivation

Problem:

As people are tied up in their occupations, they have a tend to leave their household work unattended. When clothes are kept out for drying, due to unpredictable weather conditions people struggle with uneven drying or exposure to adverse weather which can damage clothes or delay drying times.

1. Changing weather conditions:

A favourable weather condition is a reliant factor when it comes to drying clothes. Rain, humidity, strong winds can delay the drying process of clothes, and they can cause damages to the clothes as well.

2. Time consuming:

With people's busy lives, hanging and retrieving clothes can be time consuming, especially for large households or people with busy lifestyles.

Motivation:

With the implement of this home automation system, it would be enhancing convenience and efficiency in household chores.

1. Time Savings: Automation reduces the need for manual intervention, allowing users to focus on other tasks.
2. Improved Efficiency: The system will optimize drying based on environmental conditions, ensuring clothes dry faster and more evenly.

3. Environmental Impact: Efficient drying reduces energy consumption by minimizing the need for electric dryers.
4. Enhanced Convenience: Integration with smart home systems will provide users with control and monitoring capabilities from their mobile devices.

Aim and Objectives

Aim:

Our aim through this is to propose a weather-adaptive cloth line system that transforms the traditional laundry routine into a seamless, intelligent experience. This would be a cutting-edge solution that brings a fresh, smart approach to a common household task.

Objectives:

Designing the System: Developing a detailed hardware and software design for the automated cloth line system.

Selecting Components: Choosing appropriate sensors, motors, and microcontrollers for the system.

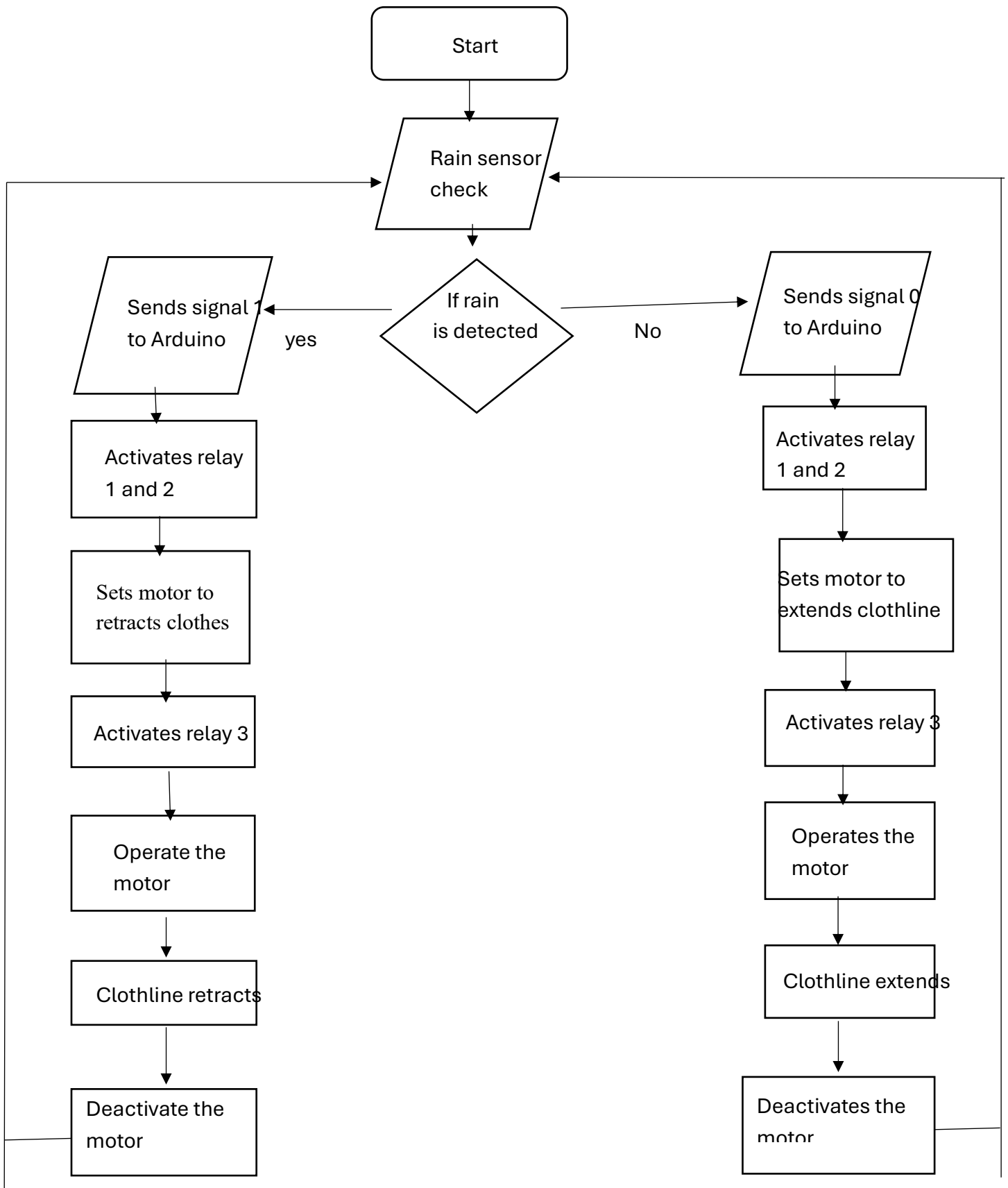
Developing the Control System: Program the Arduino microcontroller to interface with sensors and motors for automatic adjustments.

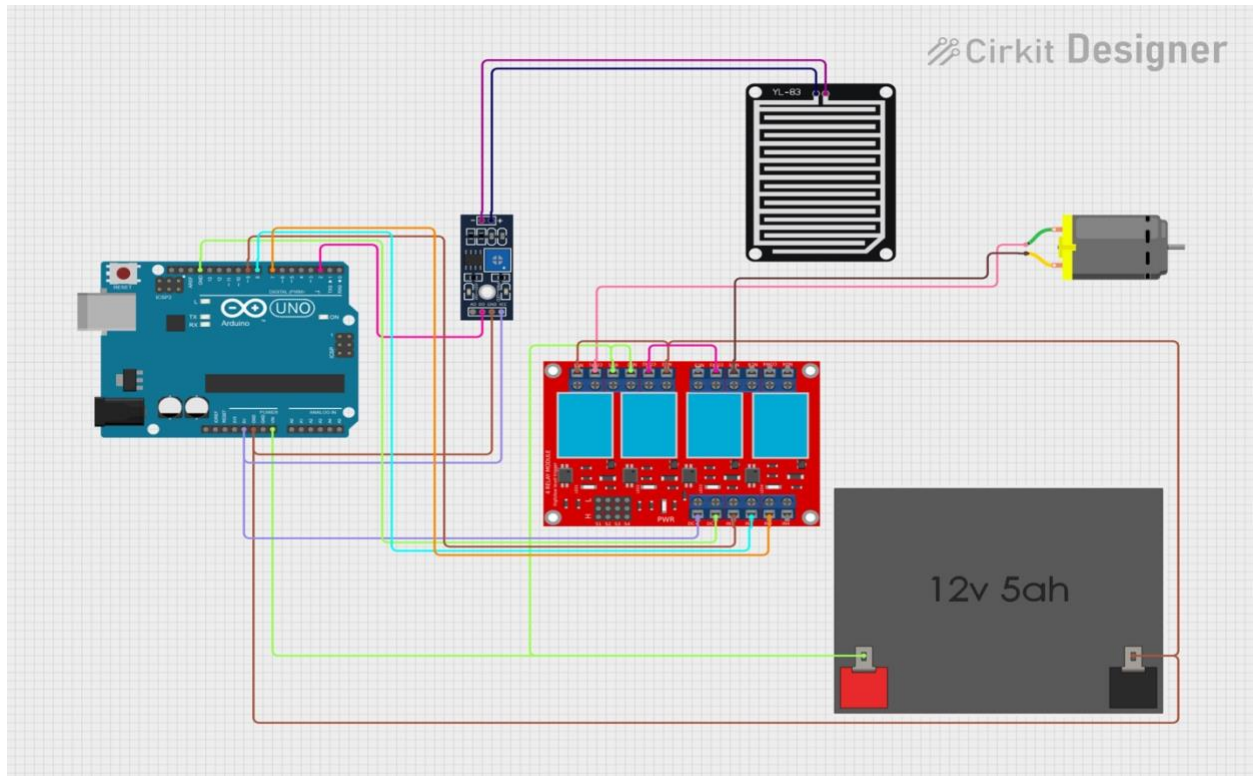
Creating the User Interface: Developing a mobile interface for remote monitoring and control of the system.

Test and Calibrate: Conduct testing to ensure the system operates reliably under various conditions and refine the calibration as needed.

Evaluate Performance: Assess the system's performance against predefined criteria to validate its effectiveness and efficiency.

System diagram



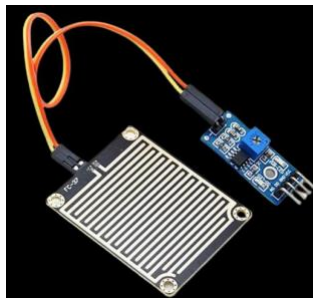


Methodology

This outlines the methodology for developing our automated cloth line system. For this we have used a rain sensor, some relays and a motor to retract the cloth line during rainfall.

Rain sensor:

Rain sensor detects the presence of rain and generates an electrical signal.



Rain sensor has two main parts:

1. Sensor pad: This is made up of copper particles on both sides. This acts as a variable resistor. When there is more rainwater, it has a good conductivity, so the resistance is low.

A less conductivity occurs when there is less rainwater, thereby, increasing the resistance. So, the resistance value is inversely proportional to the amount of water. Because of this concept we can get an output voltage

2. The module: Sensor pad should be connected to this. This has three pins which should be connected to the Arduino board. The below table explains where the pins should be connected to,

Sensor pins	Arduino pins
VCC	+5V
GND	GND
DO	Any pin from (1 to 13)

Relay:

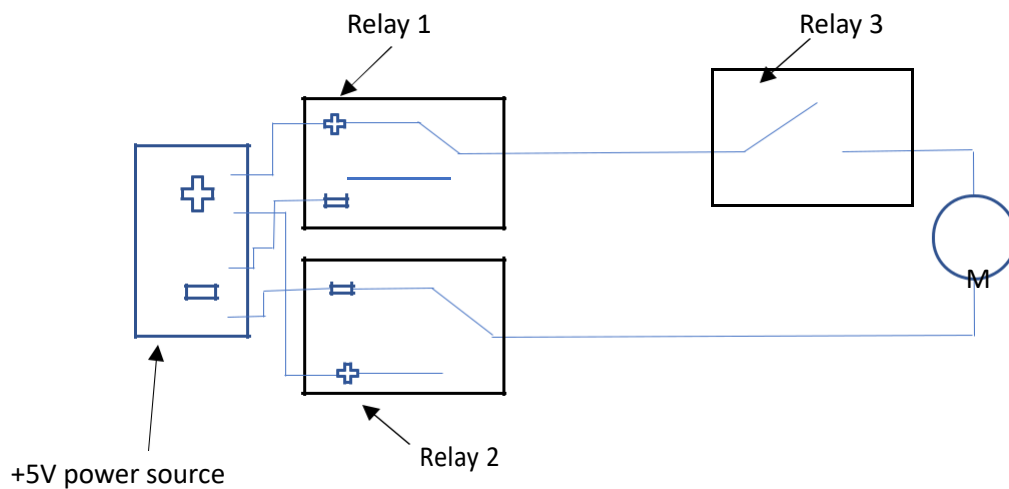
Relay acts as a switch. According to the signal given by the Arduino, relay is controlled. The relay should be connected to the Arduino as below,

Relay pins	Arduino pins
VCC	+5V
GND	GND
In 1	Any pin from (1 to 13) e.g.:- 11 pin
In 2	Any pin from (1 to 13) e.g.:- 10 pin
In 3	Any pin from (1 to 13) e.g.:- 9 pin



Motor:

Then the relay should be connected to the motor. In here we use 3 relays. The below diagram shows how to connect them.



Here,, relay 1 and relay 2 is used for polarity switching while the relay 3 is used to operate the motor for precisely the required period of time.

Evaluation Method

This topic involves the accuracy and reliability of the sensor, the effectiveness of the indicators, memory handling & system response.

Sensors: Rain Sensor

We can test the sensor by introducing slight, medium and intense raining incidence on it. Then we can understand the duration that takes the sensor to take the measurements when moisture level is changing. The response time should be such that this can be run in real-time operation fashion.

Indicators: Relays and Motor

We can control relays and the motor as per the readings received by the sensors. Then we can check that the relays and motor function as expected concerning the signals from the rain sensor for instance to operate when it is raining. Thus, making sure that the flag operates every time a sensor is detected as it has been mentioned in this assignment.

Memory Device: Arduino Board

We can test this by regulating a set of rules in Arduino that permits the sensor data or system states to be saved. Then we can understand that the Arduino can send and store data without glitches and at times also obtain a good quality reading. The response time of system can be tested using the various sensor inputs and control commands.

Error Handling and Recovery: Testing the ability of Arduino to fail over and gracefully handle errors.

References

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<https://docs.arduino.cc/tutorials/>

https://youtu.be/zJ-LqeX_fLU?si=ye-12XD9-OlmkW8k

<https://youtu.be/DPqilzK97K0?si=5VYjf1Agzh7YK1JQ>